IONIC Equilibrium

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ACID - BASE

1) Arrhenius

→ Aq. sol^n mein → H^+ / $^{\ominus}OH$

2) Bronsted - lowry

→ H^+ donor / acceptor

3) Lewis

→ e$^-$ pair donor / acceptor

* $H_2O_{(l)} \rightleftharpoons H^{\oplus} + {}^{\ominus}OH$

$$K_a = \frac{[H^+][^-OH]}{[H_2O]}$$

$$K_a[H_2O] = K_w = [H^+][^-OH]$$

at 25°C → 10^{-14}

at 63° C → 10^{-13}

* $p^H + p^{OH} = 14$

* $p^x = -\log x$

log 2	= 0.3
log 3	= 0.47
log 7	= 0.84

* Weak acid (HA) → K_a → diss. const.
c → conc^n

$$pH = \frac{1}{2}\left(p^{K_a} - \log c\right)$$

* Hydrolysis

$$K_h = \frac{K_w}{K_a}$$

Degree of hydrolysis → h (same as α)

1. Basic Salt. (W.A + S.B)

$$pH = 7 + \frac{1}{2}\left(p^{K_a} + \log c\right)$$

(K_a → of the weak acid)

2. Acidic Salt. (S.A + W.B)

$$pH = 7 + \frac{1}{2}\left(p^{K_b} + \log c\right)$$

(K_b → of weak base)

3. W.B + W.A

$$pH = 7 + \frac{1}{2}\left(p^{K_a} - p^{K_b}\right)$$

* Amphiprotic

$$pH = \frac{p^{K_{a_1}} + p^{K_{a_2}}}{2}$$

(K_{a_1} → I diss., K_{a_2} → II diss.)

* BUFFER

$$pH = pK_a + \log \frac{[\text{conjugate base}]}{[\text{weak acid}]}$$

Indicators (HIn)

$HIn \rightarrow H^{\oplus} + In^{\ominus}$

$$pH = pK_{In} + \log \frac{[In^{\ominus}]}{[HIn]}$$

HPh →	8.3 - 10
MeOH →	3.1 - 4.4
B.T.B →	6 - 7.6

(Bromo Thymol Blue)

← pH Range

Solubility (S) & Sol. Product (K_{sp})

$$A_xB_y \rightleftharpoons xA^{y+} + yB^{x-}$$

$$K_{sp} = x^x \cdot y^y \cdot s^{(x+y)}$$